

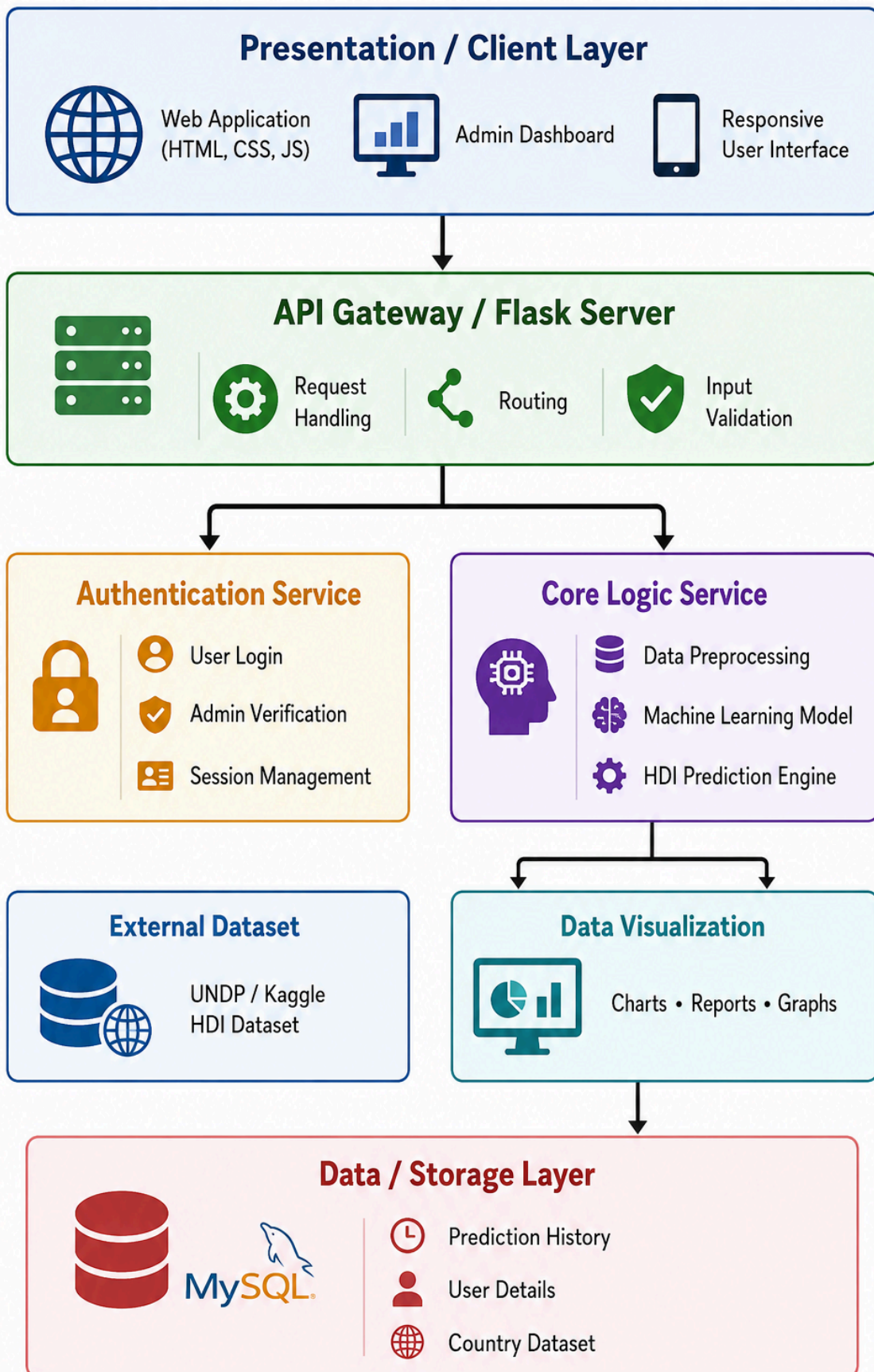
Solution Architecture

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Team ID: XXXXXX

Project Name: Human Development Index (HDI) Prediction using Machine Learning

Solution Architecture of the Human Development Index (HDI) Prediction System



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a user-friendly interface for entering country development indicators, viewing HDI predictions, dashboards, and reports.	HTML5, CSS3, Bootstrap, JavaScript
API Gateway / Server	Receives user requests, validates input data, routes requests to backend services, and returns prediction results.	Python, Flask
Authentication Service	Manages user login, administrator access, and session security.	Flask Login, Password Hashing
Core Logic Service	Performs data preprocessing, feature selection, machine learning prediction, and result generation.	Python, Scikit-learn, Pandas, NumPy
External Data Integration	Imports HDI datasets and country development data for training and prediction.	UNDP Dataset, Kaggle Dataset, CSV Files
Visualization Module	Generates interactive charts, graphs, and downloadable reports for analysis.	Matplotlib, Seaborn, Plotly
Database	Stores user information, prediction history, country datasets, and system logs.	MySQL
Deployment Platform	Hosts the web application and machine learning services for public access.	Render / Railway / Heroku

Architecture Description

The **Human Development Index (HDI) Prediction System** follows a **three-layer architecture** consisting of the **Presentation Layer**, **Application Layer**, and **Data Layer**. Users interact with the web interface to provide development indicators such as Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) per capita. The Flask server validates the input and forwards it to the machine learning prediction engine. The core logic preprocesses the data, executes the trained machine learning model, and predicts the HDI category. The prediction results are visualized through dashboards and stored in the MySQL database along with user and historical

prediction data. This layered architecture ensures scalability, maintainability, security, and efficient processing of HDI prediction requests.
